

BRIDGING THE GAP: TRANSFER LEARNING FOR ENGLISH / KISWAHILI → EKEGUSII

PSA TRANSLATION

Fine-tuning mT5-small and NLLB-200 to translate Public Service Announcements into Ekegusii — a low-resource Bantu language

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INTRODUCTION & MISSION

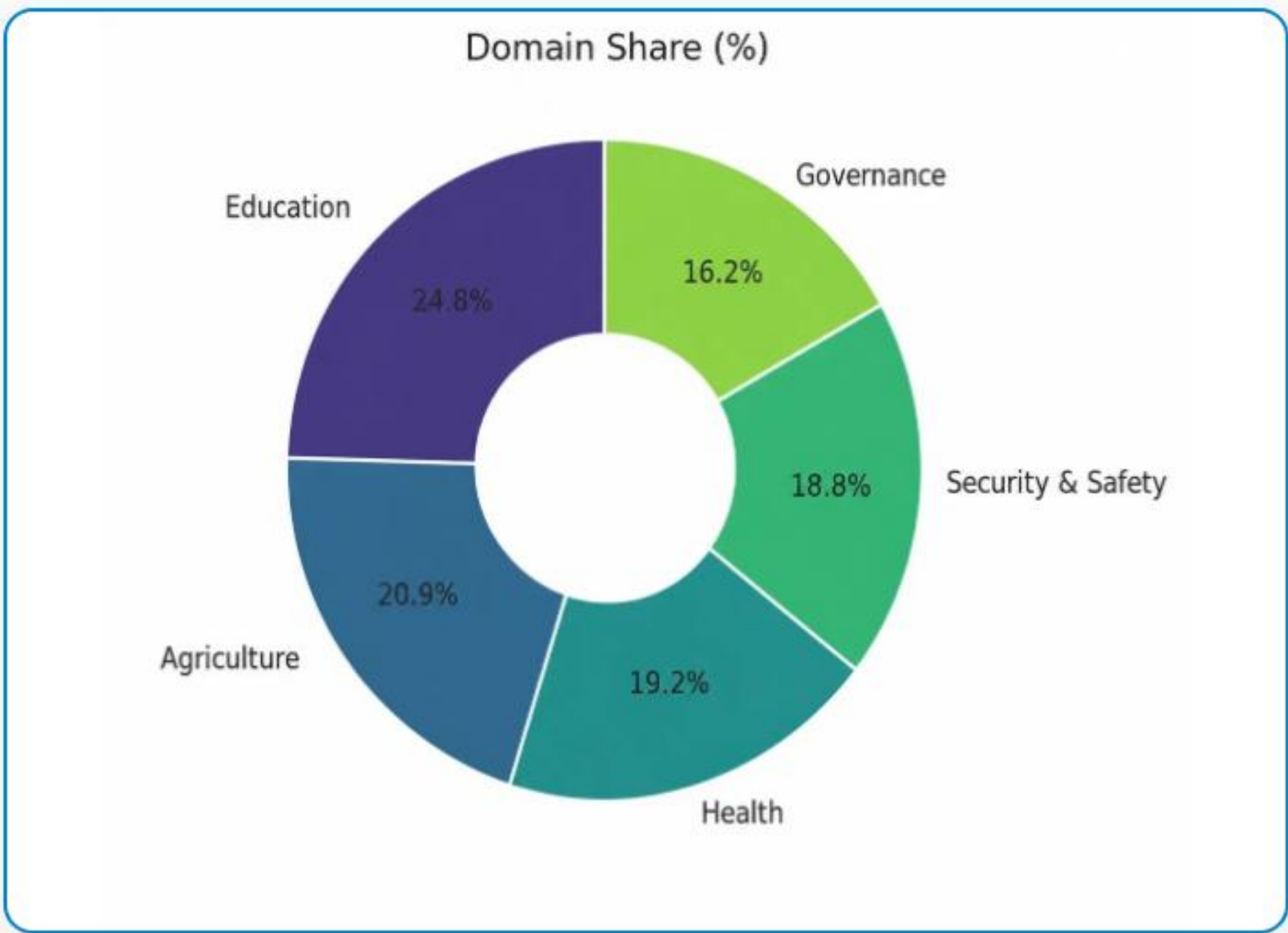
Problem: Public Service Announcements (PSAs) covering health, agriculture, civic safety, governance, and education are widely produced in English and Kiswahili, leaving over 2.2 million Ekegusii speakers with limited access to vital information.

Mission: Leverage state-of-the-art multilingual neural machine translation models (mT5-small and NLLB-200) to automate accurate translation of public notices into Ekegusii.

HYBRID DATASET & DOMAINS

Data Sources: Hybrid dataset consisting of real scraped official Kenyan government and non-government organizations publications + synthetically generated pairs using the real ones as bootstrapping (~21,000 sentence triplets).

Distribution Across All 5 PSA Domains:



- Education: 24.8%
- Agriculture: 20.9%
- Health: 19.2%
- Security & Safety: 18.8%
- Governance: 16.2%

MODEL METHODOLOGY

- mT5-small:** Conditioned with explicit task prefixes (e.g., "translate English to Ekegusii:"), freezing bottom 4 of 8 encoder layers.
- NLLB-200-distilled-600M:** Configured with placeholder target language tags (sw_h_Latn), freezing bottom 6 of 12 encoder layers.
- Data Splitting:** Grouped strictly by PSA_ID (GroupShuffleSplit) to prevent data leakage of identical target sentences across train, validation, and test splits.

EVOLUTION ACROSS EPOCHS

Training progression log across 5 fine-tuning epochs

mT5-small Training Epoch Log:

Loading weights: 100% [00:00:00.00] 192/192 [00:00:00.00, 26900.93it/s] [transformers] The tied weights mapping and config for this model specifies to tie shared weights with different values, so we will NOT tie them. You should update the config with 'tie_weights': 'none' to avoid this warning. [5330/5330 1:29:56, Epoch 5/5]				
Epoch	Training Loss	Validation Loss	Bleu	Chrf
1	4.040248	3.720722	1.712680	21.079013
2	3.732924	3.507565	2.386553	22.578745
3	3.572714	3.414613	2.620530	23.929362
4	3.525028	3.365422	2.985632	24.637274
5	3.495724	3.347350	3.112052	25.333702

Writing model shards: 100% [00:01:00.00] 1/1 [00:01:00.00, 1.65s/it]

Writing model shards: 100% [00:01:00.00] 1/1 [00:01:00.00, 1.60s/it]

Writing model shards: 100% [00:01:00.00] 1/1 [00:01:00.00, 1.70s/it]

Writing model shards: 100% [00:01:00.00] 1/1 [00:01:00.00, 1.59s/it]

Writing model shards: 100% [00:01:00.00] 1/1 [00:01:00.00, 1.61s/it]

[transformers] There were missing keys in the checkpoint model loaded: ['encoder.embed_tokens.weight']
mT5 training time: 89.9 minutes

Writing model shards: 100% [00:01:00.00] 1/1 [00:01:00.00, 1.59s/it]

NLLB-200 Training Epoch Log:

Loading weights: 100% [00:00:00.00] 512/512 [00:00:00.00, 27908.25it/s] [10655/10655 1:32:26, Epoch 5/5]				
Epoch	Training Loss	Validation Loss	Bleu	Chrf
1	3.290284	3.252705	4.391404	30.026409
2	3.075516	3.119753	5.322780	31.886891
3	2.852831	3.072481	6.068240	33.488015
4	2.790013	3.061522	6.198036	33.773248
5	2.724451	3.060054	6.379174	34.106678

Writing model shards: 100% [00:03:00.00] 1/1 [00:03:00.00, 3.22s/it]

Writing model shards: 100% [00:03:00.00] 1/1 [00:03:00.00, 3.15s/it]

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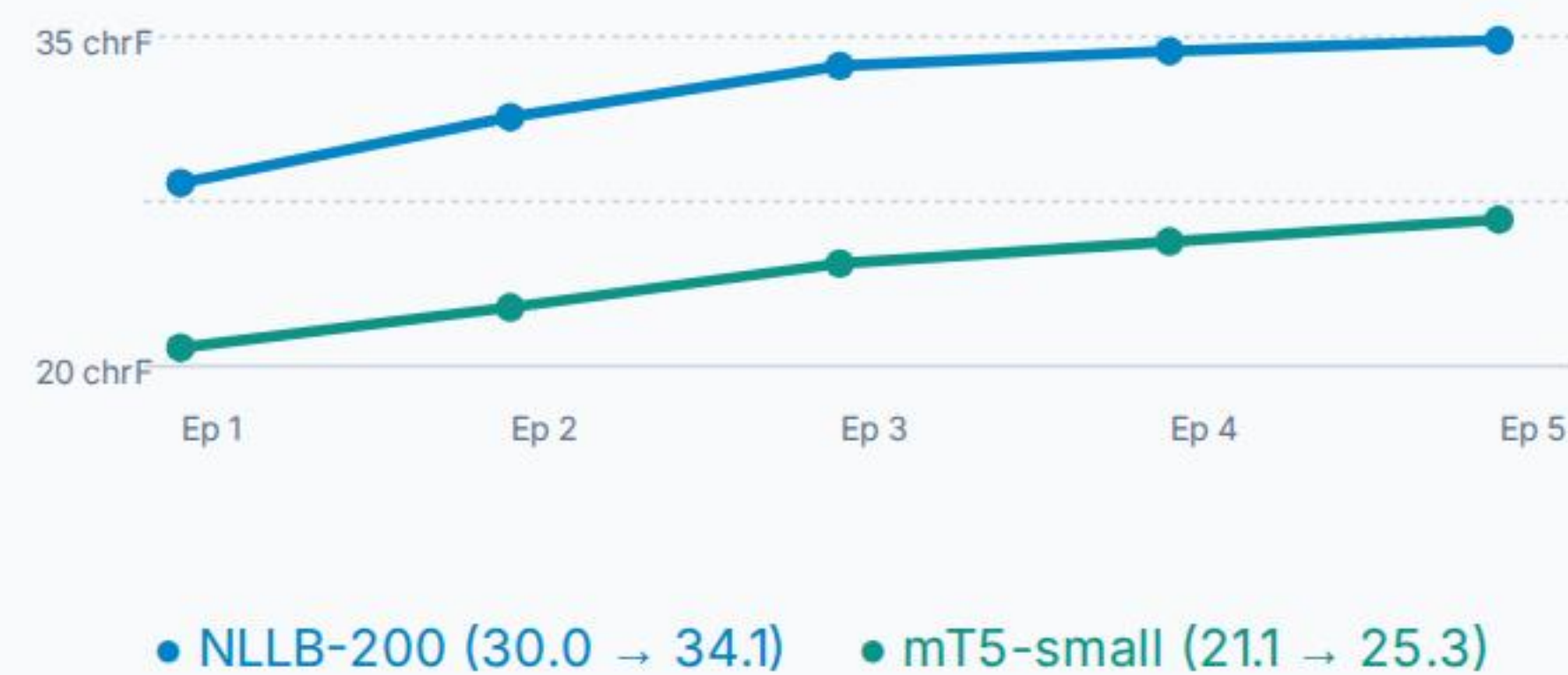
Writing model shards: 100% [00:03:00.00] 1/1 [00:03:00.00, 3.36s/it]

Writing model shards: 100% [00:03:00.00] 1/1 [00:03:00.00, 3.36s/it]

[transformers] There were missing keys in the checkpoint model loaded: ['model.encoder.d_weight']
NLLB training time: 92.4 minutes

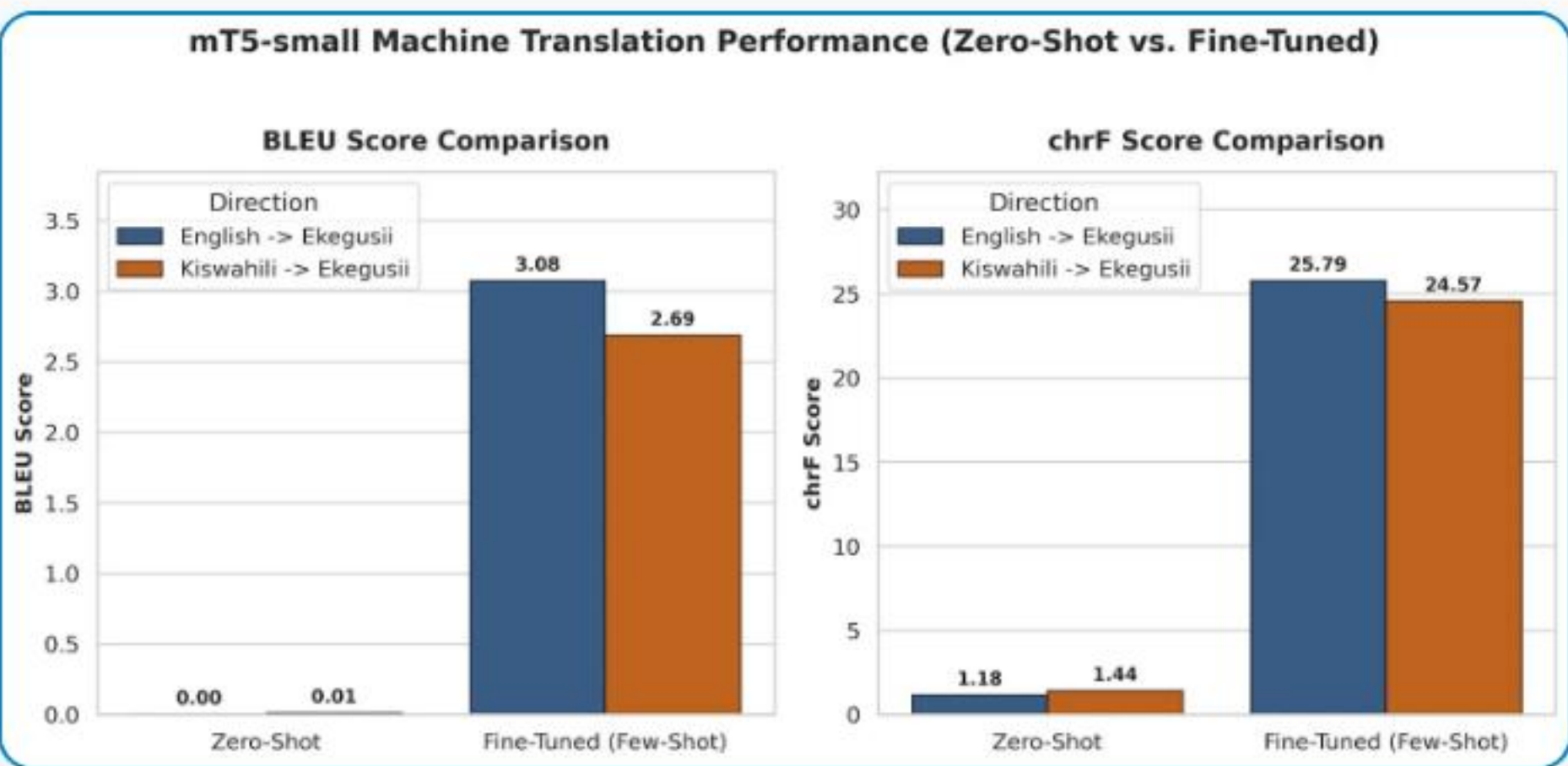
Writing model shards: 100% [00:03:00.00] 1/1 [00:03:00.00, 3.20s/it]

Validation chrF Progress Comparison:

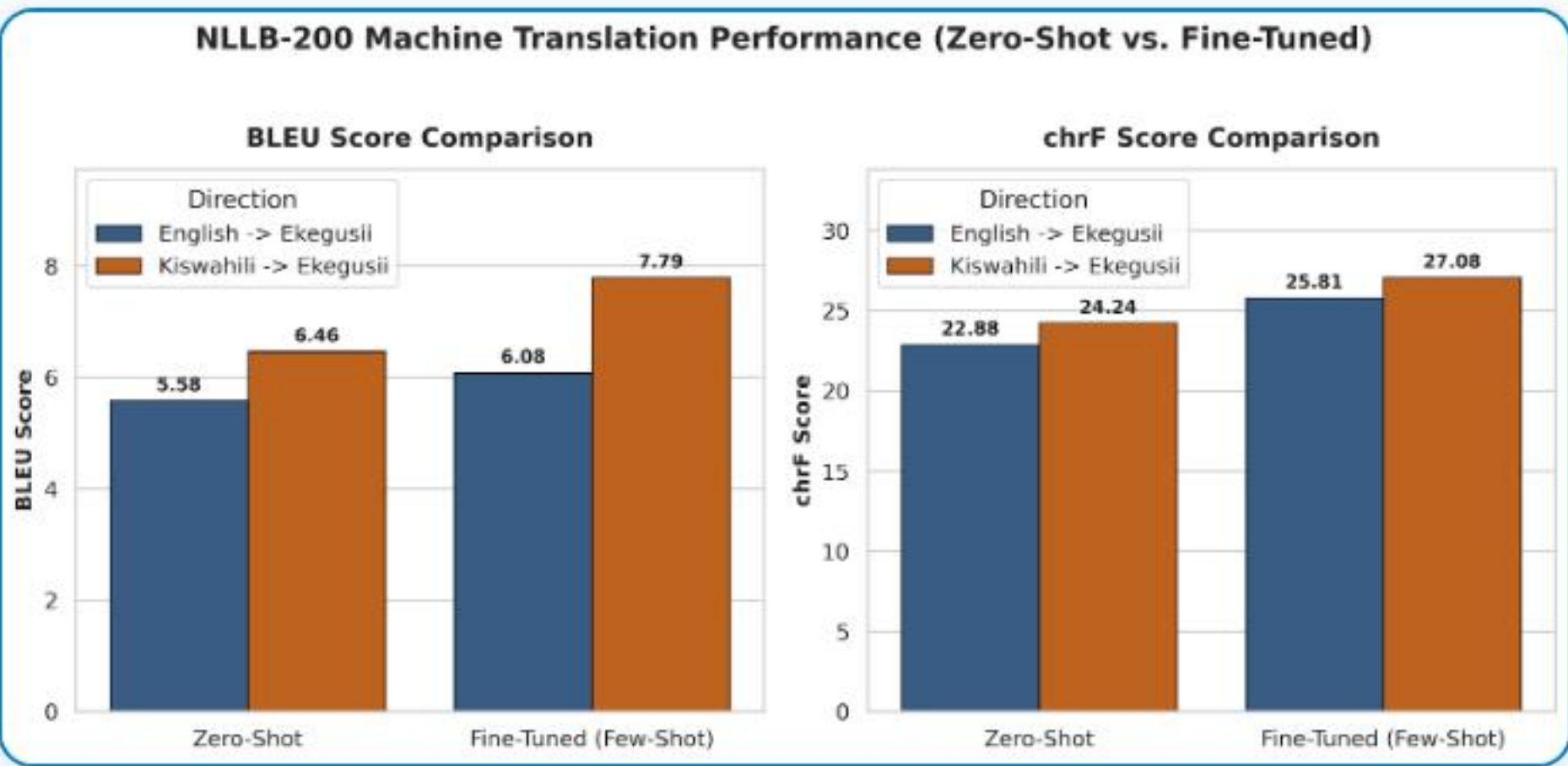


EXPERIMENTAL RESULTS & METRICS

mT5-small Evaluation Results:



NLLB-200 Evaluation Results:



QUALITATIVE TRANSLATION SAMPLES

English Source Outputs:

SRC: Farmers are urged to use agrochemicals safely.
OUT: Abaremi baborigwe gotumia ebimeri bia agrochimika buya.

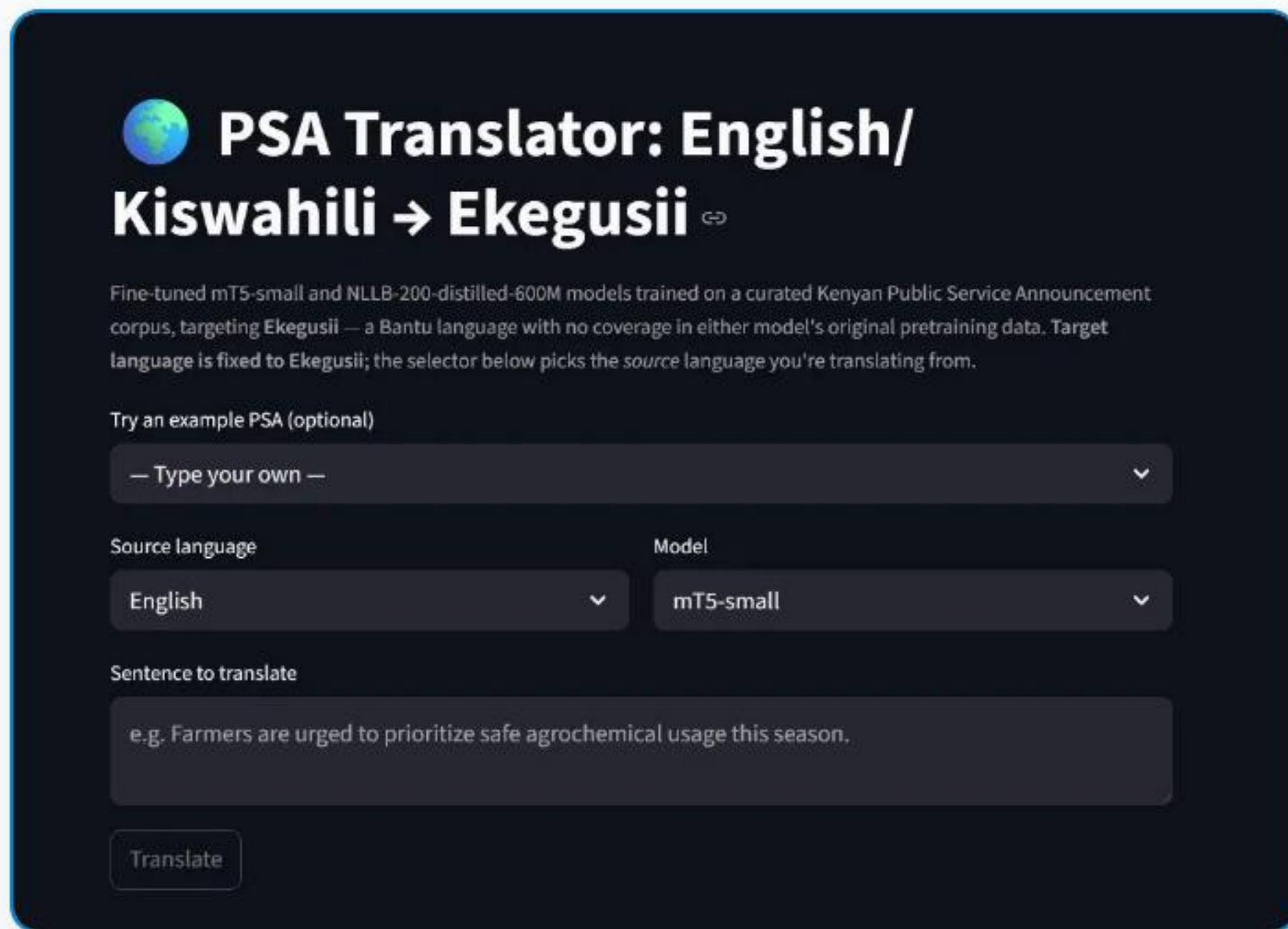
SRC: Good morning news.
OUT: Okoraria igoro y'ebirengererio.

Kiswahili Source Outputs:

Loaded on GPU
SRC: Wakulima wanahimizwa kutumia dawa za kilimo kwa usalama.
OUT: Abaremi bamenyete gosegeta ebirengererio bia oboremi bwoboremi.

SRC: Habari za asubuhi.
OUT: Omoroberio bwechibesa chiobobundi nechisemi chiokwegendereria.

STREAMLIT DEPLOYED APPLICATION



Interactive web application enabling real-time PSA translation into Ekegusii with confidence score logging.

CHALLENGES & LIMITATIONS

- Manual Verification:** It was challenging since it required manual native Ekegusii speakers to verify translation quality and validation of final outputs.
- Compute Constraints:** Computation power was a challenge, not enabling us to fine-tune, rerun, and fix again.
- Model Accuracy & Performance:** Regarding model performance, translations were not accurate enough and need more work before full field deployment.
- Language Interference:** NLLB-200 occasionally produced fluent Kiswahili instead of Ekegusii due to its strong pretrained Kiswahili associations and reliance on placeholder language tags.

FUTURE WORK & EXTENSIONS

- Dedicated Language Tokens:** Extend NLLB's vocabulary with native Ekegusii special tokens instead of reused placeholders.
- Corpus Expansion:** Expand training samples evenly across all 5 domains (Education, Agriculture, Health, Security & Safety, Governance).
- Human & COMET Evaluation:** Integrate COMET scoring alongside structured native-speaker adequacy and fluency assessments.